



CHEMISTRY CH.14 — COMMON NAMES & FORMULAS — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway



Topics in this chapter:

1. Sodas: Caustic, Baking & Washing
2. Calcium Compounds Family
3. Vitriols — Colour-based Names
4. Gemstones & Their Composition
5. Common Names of Metals & Gases
6. IUPAC Names & Organic Compounds
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9. Food Additives & Preservatives
10. ⚠ Traps / Key Differences Summary

1. Sodas: Caustic, Baking & Washing

- Caustic Soda: NaOH (Sodium Hydroxide).
 - Uses: Cleansing agent, washing soda manufacture, drain cleaner, soap/detergent, paper & pulp, fabric industry, aluminium extraction (bauxite purification). [Q1527, Q1541]
- Baking Soda: NaHCO_3 (Sodium Bicarbonate / Sodium Hydrogen Carbonate).
 - Preparation: $\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 + \text{NH}_3 \rightarrow \text{NaHCO}_3 + \text{NH}_4\text{Cl}$.
 - Uses: Antacid, fire extinguisher, baking cakes/bread, bee sting relief, milk preservation. [Q1527, Q1535, Q1536, Q1554]
- Washing Soda: $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ (Sodium Carbonate Decahydrate).
 - Uses: Glass, soap, paper industries; removing permanent hardness; stain removal; textiles; petroleum refining. [Q1527, Q1554]
- Baking Powder: Mixture of NaHCO_3 + Tartaric Acid + cornstarch buffer. [Q1535, Q1554]

2. Calcium Compounds Family

- Marble / Limestone / Chalk: CaCO_3 (Calcium Carbonate).
 - Formation: $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$. [Q1528, Q1551, Q1554]
- Quicklime: CaO (Calcium Oxide).
 - Preparation: $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ (heating limestone).
 - Reacts vigorously with water $\rightarrow$ Slaked lime + heat. [Q1547, Q1551, Q1554]
- Slaked Lime: Ca(OH)_2 (Calcium Hydroxide).
 - Other names: Hydrated lime, Lime water, Pickling lime.
 - Uses: Mortar, plaster, cement, water treatment, sewage treatment, paper production. [Q1552, Q1554]
- Gypsum: $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (Calcium Sulphate Dihydrate).
 - Uses: Wallboard, ceiling, cement retarder, fertilizer, paper filler. [Q1532, Q1554]
- Plaster of Paris: $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ (Calcium Sulphate Hemihydrate).
 - Preparation: Gypsum ko 373 K par heat karke. [Q1554]
- Calcium Phosphate: $\text{Ca}_3(\text{PO}_4)_2$ (Tricalcium Phosphate).

- Preparation: $3\text{Ca}(\text{OH})_2 + 2\text{H}_3\text{PO}_4 \rightarrow \text{Ca}_3(\text{PO}_4)_2 + 6\text{H}_2\text{O}$.
- Insoluble in ethanol/acetic acid; soluble in dilute HNO_3/HCl . [Q1559]
- Calcium Carbide: CaC_2 — acetylene production. [Q1547]
- Calcium Bicarbonate: $\text{Ca}(\text{HCO}_3)_2$ — food additive, anti-caking agent. [Q1528]
- Calcium Chloride: CaCl_2 — drying agent, highly soluble. [Q1552]
- Calcium Oxide as Drying Agent: Ammonia gas dry karne ke liye use hota hai. [Q1547]

3. Vitriols — Colour-based Names

- Blue Vitriol: $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (Hydrated Copper Sulphate) — fungicide, battery electrolytes. [Q1530, Q1532]
- Green Vitriol: $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ (Ferrous Sulphate / Iron(II) Sulphate) — iron deficiency medicine. [Q1530, Q1549]
- White Vitriol: $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ (Zinc Sulphate Heptahydrate). [Q1530, Q1532, Q1533]
- Red Vitriol: CoSO_4 (Cobalt Sulphate). [Q1530]
- Vitriol of Mars: $\text{Fe}_2(\text{SO}_4)_3$ (Ferric Sulphate). [Q1530]
- Spirit of Vitriol: H_2SO_4 (Sulphuric Acid) — also called Oil of Vitriol / Mattling acid. [Q1530, Q1542]
- Oil of Vitriol: Same as Sulphuric acid (H_2SO_4). [Q1542]

4. Gemstones & Their Composition

- Ruby: Al_2O_3 (Aluminium Oxide) — pink to blood-red. [Q1531]
- Diamond: C (Carbon). [Q1531]
- Topaz: $\text{Al}_2\text{SiO}_4(\text{F},\text{OH})_2$. [Q1531]
- Quartz: SiO_2 . [Q1531]
- Emerald: $\text{Be}_3\text{Al}_2(\text{SiO}_3)_6$. [Q1531]
- Opal: $\text{SiO}_2 \cdot n\text{H}_2\text{O}$. [Q1531]

5. Common Names of Metals & Gases

- Mercury (Hg): Common name = Quicksilver; atomic number 80; d-block; Group 12. [Q1533, Q1555]
 - Uses: Thermometers, barometers, manometers, fluorescent lighting, switches.
- Laughing Gas: N_2O (Nitrous Oxide). [Q1534]
- Dry Ice: Solid CO_2 — sublimates directly to gas without melting. [Q1558]
- Rock Salt: Mineral name = Halite; mainly NaCl. [Q1529]
- Rust: $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ (Hydrated Iron(III) Oxide). [Q1546]
 - Reaction: $\text{Iron} + \text{O}_2 + \text{Water} \rightarrow \text{Iron Oxide (Rust)}$.

6. IUPAC Names & Organic Compounds

- Propane: $\text{CH}_3\text{CH}_2\text{CH}_3$ (C_3H_8) — 3 carbon alkane. [Q1538]
- Propanone: CH_3COCH_3 (Acetone / Dimethyl ketone) — ketone group ($-\text{CO}-$) with 3 carbons. [Q1539]
- Ethanoic Acid = Acetic Acid (CH_3COOH) — carboxylic acid group. [Q1542]

- Uses: Photographic film, wood glue, synthetic fibres, pickled vegetables, vinegar.
- Ethane (C_2H_6): Total 7 covalent bonds (1 C-C + 6 C-H). [Q1540]
- Alkanes: C_nH_{2n+2} (single bonds) — CH_4 , C_2H_6 , C_4H_{10} . [Q1538]
- Alkenes: C_nH_{2n} (at least one double bond) — more reactive than alkanes. [Q1538]
- Alkynes: C_nH_{2n-2} (triple bond). [Q1538]
- Covalent Bond Types:
 - Single: H_2 , Cl_2
 - Double: O_2 , CO_2
 - Triple: N_2 . [Q1540]
- IUPAC (International Union of Pure and Applied Chemistry): World authority on chemical nomenclature aur terminology — elements, symbols, units, naye elements ke naam sabki approval karta hai. 1919 mein establish hua tha; headquarters Paris, France mein hai; International Science Council (ISC) ka member hai. [Q1537]

7. Important Chemical Formulas

- Caustic Potash: KOH (Potassium Hydroxide) — food thickener, stabilizer, potassium soaps. [Q1541]
- Milk of Magnesia: $Mg(OH)_2$ (Magnesium Hydroxide) — antacid, white solid, low water solubility. [Q1543]
- Ammonium Hydroxide: NH_4OH (Ammonia water) — mildly acidic solution. [Q1541, Q1543]
- Aluminium Sulphate: $Al_2(SO_4)_3$ — coagulating agent in water purification, paper manufacturing. [Q1545]
- Lithium Nitride: Li_3N — only stable alkali metal nitride; Li (+1), N (-3). [Q1548]
- Sodium Sulphate: Na_2SO_4 . [Q1553]
- Bleaching Powder: $CaOCl_2$ (Calcium Hypochlorite) — oxidising agent, water disinfectant. [Q1554]
- Ammonium Phosphate: $(NH_4)_3PO_4$ — fertilizer ingredient. [Q1556]
- Lactose (Milk Sugar): $C_{12}H_{22}O_{11}$ — disaccharide of β -D-galactose + β -D-glucose. [Q1557]
 - Glucose/Galactose/Fructose = $C_6H_{12}O_6$ (monosaccharides). [Q1557]
- Monosodium Glutamate (MSG): "Sugar salt" — flavour enhancer in Chinese dishes. [Q1550]

8. Latin Names of Elements

- Sodium (Na): Natrium. [Q1544, Q1560]
- Silver (Ag): Argentum. [Q1560]
- Potassium (K): Kalium. [Q1544]
- Sulphur (S): Sulfurium. [Q1544]
- Nitrogen (N): Nitrum. [Q1544]
- Other Latin-derived symbols: Copper (Cu), Gold (Au), Lead (Pb), Iron (Fe), Tin (Sn), Mercury (Hg). [Q1560]

9. Food Additives & Preservatives

- Monosodium Glutamate (MSG): Flavour enhancer; restaurant foods, canned vegetables, soups. [Q1550]
- Sodium Benzoate: Food preservative. [Q1550]
- Calcium Carbonate: Chalk/limestone — NOT sugar salt. [Q1550]
- Calcium Hypochlorite: Bleaching agent & disinfectant — NOT sugar salt. [Q1550]

10. ⚠ Traps / Key Differences Summary

- ⚠ Trap: Caustic soda = NaOH; Caustic potash = KOH — dono alag hain. [Q1527, Q1541]
- ⚠ Trap: Baking soda = NaHCO_3 ; Baking powder = NaHCO_3 + Tartaric acid — dono alag hain. [Q1535, Q1554]
- ⚠ Trap: Ethanoic acid = Acetic acid — same compound, do naam. [Q1542]
- ⚠ Trap: Propane ($\text{CH}_3\text{CH}_2\text{CH}_3$) aur Propanone (CH_3COCH_3) alag hain — ketone group pehchano. [Q1538, Q1539]
- ⚠ Trap: Ethane (C_2H_6) mein total 7 covalent bonds hain — 1 C-C + 6 C-H. [Q1540]
- ⚠ Trap: Quicklime = CaO; Slaked lime = $\text{Ca}(\text{OH})_2$ — dono alag hain. [Q1551, Q1552]
- ⚠ Trap: Quicksilver = Mercury (Hg); Silver = Argentum (Ag) — dono alag hain. [Q1533, Q1555]
- ⚠ Trap: Dry ice = Solid CO_2 — ye ice nahi hai, sublimation se gas ban jaata hai. [Q1558]
- ⚠ Trap: Rock salt ka mineral name Halite hai — Limonite/Hematite/Siderite nahi. [Q1529]
- ⚠ Trap: Ruby = Al_2O_3 (Aluminium oxide) — Sodium chloride/Carbon/Potassium nahi. [Q1531]
- ⚠ Trap: Lactose = $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ — Glucose/Galactose/Fructose sab $\text{C}_6\text{H}_{12}\text{O}_6$ hain. [Q1557]
- ⚠ Trap: Rust = $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ — FeO ya Fe_2O nahi. [Q1546]
- ⚠ Trap: Natrium = Sodium aur Argentum = Silver — dono statements true hain. [Q1560]
- ⚠ Trap: Element names/symbols/units ki approval IUPAC karta hai — koi aur international body nahi. [Q1537]